

Supplementary tables for the article “Model-comparison effect size indices for logistic and other generalized linear models: Computation and power analysis”

Simulation results tables

The figures in the main text plot the results of Study 1 (recovery of the whole-model R^2 and of a single focal predictor’s eta-squared) and Study 2 (simulation-based required sample size). For Study 1, each Model x N x k combination in the figures is a curve over 10 population effect size values; the tables below summarize accuracy and power collapsed across those population effect size values only (mean signed bias, mean absolute bias, RMSE, and mean power gap, by model, sample size, and number of covariates k), keeping the k = 3 vs. k = 5 conditions separate rather than averaging over them, so the full crossed design is preserved – the full per-point results are available in this repository’s `simulations/Data/` folder. For Study 2, which has no sample-size or k dimension, the tables report the full underlying data.

Study 1a: accuracy of R^2 (collapsed summary of Figure 1)

Table 1: Accuracy of R^2 and adjusted R^2 , by model, N, and number of covariates k, collapsed across population R^2 (data underlying Figure 1). Bias columns report the mean signed bias, so the direction of the deviation is preserved.

Model	N	k	Bias (R^2)	Mean Bias (R^2)	RMSE (R^2)	Bias (Adjusted R^2)	Mean Bias (Adjusted R^2)	RMSE
Logistic	30	3	0.086	0.086	0.155	0.012	0.012	
Logistic	30	5	0.144	0.144	0.200	0.021	0.021	
Logistic	60	3	0.041	0.041	0.093	0.004	0.004	
Logistic	60	5	0.068	0.068	0.112	0.007	0.007	
Logistic	90	3	0.027	0.027	0.071	0.003	0.003	
Logistic	90	5	0.044	0.044	0.081	0.003	0.003	
Logistic	120	3	0.020	0.020	0.059	0.002	0.002	
Logistic	120	5	0.032	0.032	0.066	0.002	0.002	
Multinomial	30	3	0.108	0.108	0.147	0.014	0.015	
Multinomial	30	5	0.176	0.176	0.205	0.018	0.023	
Multinomial	60	3	0.053	0.053	0.083	0.006	0.006	
Multinomial	60	5	0.088	0.088	0.112	0.010	0.010	
Multinomial	90	3	0.034	0.034	0.060	0.003	0.003	
Multinomial	90	5	0.057	0.057	0.078	0.005	0.005	
Multinomial	120	3	0.025	0.025	0.049	0.002	0.002	
Multinomial	120	5	0.041	0.041	0.061	0.003	0.003	
Ordinal	30	3	0.061	0.061	0.117	0.013	0.013	
Ordinal	30	5	0.101	0.101	0.149	0.022	0.022	
Ordinal	60	3	0.029	0.029	0.071	0.005	0.005	
Ordinal	60	5	0.046	0.046	0.081	0.007	0.007	
Ordinal	90	3	0.018	0.018	0.054	0.003	0.003	
Ordinal	90	5	0.030	0.030	0.061	0.004	0.004	
Ordinal	120	3	0.014	0.014	0.046	0.002	0.002	
Ordinal	120	5	0.022	0.022	0.050	0.002	0.002	
Gaussian	30	3	0.080	0.080	0.145	-0.007	0.007	

Table 1: Accuracy of R2 and adjusted R2, by model, N, and number of covariates k, collapsed across population R2 (data underlying Figure 1). Bias columns report the mean signed bias, so the direction of the deviation is preserved. *(continued)*

Model	N	k	Bias (R2)	Mean Bias (R2)	RMSE (R2)	Bias (Adjusted R2)	Mean Bias (Adjusted R2)	RMSE
Gaussian	30	5	0.136	0.136	0.185	-0.009	0.009	
Gaussian	60	3	0.038	0.038	0.093	-0.004	0.004	
Gaussian	60	5	0.067	0.067	0.110	-0.005	0.005	
Gaussian	90	3	0.026	0.026	0.073	-0.003	0.003	
Gaussian	90	5	0.045	0.045	0.083	-0.003	0.003	
Gaussian	120	3	0.019	0.019	0.061	-0.002	0.002	
Gaussian	120	5	0.033	0.033	0.068	-0.003	0.003	

Study 1a: power of the omnibus test (collapsed summary of Figure 3)

Table 2: Mean |simulated - closed-form| power for R2 and adjusted R2, by model, N, and number of covariates k, collapsed across population R2 (data underlying Figure 3).

Model	N	k	Mean Power Gap (R2)	Mean Power Gap (Adjusted R2)
Logistic	30	3	0.151	0.015
Logistic	30	5	0.235	0.016
Logistic	60	3	0.079	0.010
Logistic	60	5	0.131	0.010
Logistic	90	3	0.051	0.006
Logistic	90	5	0.082	0.007
Logistic	120	3	0.035	0.003
Logistic	120	5	0.059	0.003
Multinomial	30	3	0.238	0.068
Multinomial	30	5	0.315	0.078
Multinomial	60	3	0.118	0.037
Multinomial	60	5	0.163	0.050
Multinomial	90	3	0.070	0.025
Multinomial	90	5	0.099	0.031
Multinomial	120	3	0.049	0.019
Multinomial	120	5	0.070	0.025
Ordinal	30	3	0.107	0.006
Ordinal	30	5	0.164	0.009
Ordinal	60	3	0.049	0.003
Ordinal	60	5	0.078	0.004
Ordinal	90	3	0.029	0.002
Ordinal	90	5	0.048	0.002
Ordinal	120	3	0.018	0.001
Ordinal	120	5	0.033	0.001
Gaussian	30	3	0.218	0.006
Gaussian	30	5	0.332	0.007
Gaussian	60	3	0.126	0.005
Gaussian	60	5	0.206	0.008
Gaussian	90	3	0.083	0.004
Gaussian	90	5	0.130	0.006
Gaussian	120	3	0.057	0.001
Gaussian	120	5	0.096	0.002

Study 1b: accuracy of eta-squared (collapsed summary of Figure 2)

Table 3: Accuracy of eta-squared and adjusted epsilon-squared, by model, N, and number of covariates k, collapsed across population eta-squared (data underlying Figure 2). Bias columns report the mean signed bias, so the direction of the deviation is preserved. Because the sign of the epsilon-squared bias reverses across effect sizes for some models (see main text), collapsed signed values can be close to zero; per-effect-size values are available in the repository’s simulations/Data folder.

Model	N	k	Bias (Eta2)	Mean Bias (Eta2)	RMSE (Eta2)	Bias (Epsilon2)	Mean Bias (Epsilon2)	RMSE (Epsilon2)
Logistic	30	3	0.032	0.032	0.125	0.009	0.009	
Logistic	30	5	0.028	0.028	0.125	0.006	0.009	
Logistic	60	3	0.015	0.015	0.082	0.003	0.003	
Logistic	60	5	0.016	0.016	0.085	0.004	0.004	
Logistic	90	3	0.010	0.010	0.065	0.002	0.002	
Logistic	90	5	0.010	0.010	0.066	0.002	0.002	
Logistic	120	3	0.007	0.007	0.055	0.001	0.001	
Logistic	120	5	0.007	0.007	0.056	0.001	0.001	
Multinomial	30	3	0.034	0.034	0.095	0.003	0.010	
Multinomial	30	5	0.025	0.029	0.092	-0.006	0.018	
Multinomial	60	3	0.020	0.020	0.064	0.004	0.005	
Multinomial	60	5	0.019	0.019	0.065	0.003	0.004	
Multinomial	90	3	0.012	0.012	0.050	0.002	0.002	
Multinomial	90	5	0.012	0.012	0.051	0.002	0.003	
Multinomial	120	3	0.008	0.008	0.042	0.001	0.001	
Multinomial	120	5	0.009	0.009	0.043	0.001	0.002	
Ordinal	30	3	0.026	0.026	0.100	0.010	0.010	
Ordinal	30	5	0.025	0.025	0.104	0.010	0.010	
Ordinal	60	3	0.012	0.012	0.065	0.004	0.004	
Ordinal	60	5	0.012	0.012	0.067	0.005	0.005	
Ordinal	90	3	0.008	0.008	0.051	0.003	0.003	
Ordinal	90	5	0.009	0.009	0.053	0.004	0.004	
Ordinal	120	3	0.006	0.006	0.044	0.002	0.002	
Ordinal	120	5	0.006	0.006	0.045	0.002	0.002	
Gaussian	30	3	0.009	0.013	0.112	-0.016	0.017	
Gaussian	30	5	-0.002	0.016	0.109	-0.026	0.027	
Gaussian	60	3	0.004	0.007	0.080	-0.009	0.009	
Gaussian	60	5	-0.001	0.008	0.078	-0.014	0.014	
Gaussian	90	3	0.003	0.004	0.065	-0.006	0.006	
Gaussian	90	5	-0.001	0.006	0.064	-0.010	0.010	
Gaussian	120	3	0.002	0.003	0.056	-0.005	0.005	
Gaussian	120	5	-0.001	0.004	0.056	-0.007	0.007	

Study 1b: power of the single-predictor test (collapsed summary of Figure 4)

Table 4: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, by model, N, and number of covariates k, collapsed across population eta-squared (data underlying Figure 4).

Model	N	k	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Logistic	30	3	0.058	0.016
Logistic	30	5	0.064	0.018
Logistic	60	3	0.027	0.005

Table 4: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, by model, N, and number of covariates k, collapsed across population eta-squared (data underlying Figure 4). *(continued)*

Model	N	k	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Logistic	60	5	0.032	0.008
Logistic	90	3	0.016	0.003
Logistic	90	5	0.017	0.004
Logistic	120	3	0.011	0.002
Logistic	120	5	0.012	0.001
Multinomial	30	3	0.069	0.019
Multinomial	30	5	0.074	0.020
Multinomial	60	3	0.032	0.006
Multinomial	60	5	0.034	0.005
Multinomial	90	3	0.020	0.002
Multinomial	90	5	0.019	0.001
Multinomial	120	3	0.011	0.001
Multinomial	120	5	0.011	0.001
Ordinal	30	3	0.044	0.011
Ordinal	30	5	0.052	0.019
Ordinal	60	3	0.017	0.002
Ordinal	60	5	0.017	0.003
Ordinal	90	3	0.009	0.001
Ordinal	90	5	0.009	0.001
Ordinal	120	3	0.005	0.001
Ordinal	120	5	0.006	0.001
Gaussian	30	3	0.049	0.033
Gaussian	30	5	0.043	0.063
Gaussian	60	3	0.032	0.009
Gaussian	60	5	0.024	0.020
Gaussian	90	3	0.022	0.004
Gaussian	90	5	0.019	0.008
Gaussian	120	3	0.015	0.002
Gaussian	120	5	0.012	0.006

Supplementary simulation: nonzero effects for the other covariates

In the η^2 simulations reported in the main text (Study 1b), the focal predictor was the only covariate with a nonzero population coefficient; the remaining $k - 1$ covariates were correlated with it ($r = .30$) but had no true effect on the outcome. To check that this design choice does not flatter the indices – with the nuisance coefficients at zero, the reduced model used to compute η^2 is close to the null model, so the deviance comparison underlying the index is not exercised in its more general, nontrivial form – Study 1b was rerun with the $k - 1$ nuisance covariates given a real, nonzero effect: half the magnitude of the focal predictor’s coefficient: `Rsimcity::sample_by_eta2()` rescales the whole coefficient vector to hit the target η^2 for the focal predictor exactly, so this pattern preserves the intended population η^2 while making the covariates relevant to the outcome. Models, sample sizes, population η^2 values, covariate correlation, exclusion criteria, and the 5,000 replications per cell are identical to Study 1b; Since number of covariates did not influence the results, only $k = 3$ was run in this supplementary simulation.

Accuracy of eta-squared and epsilon-squared, with and without nonzero nuisance effects

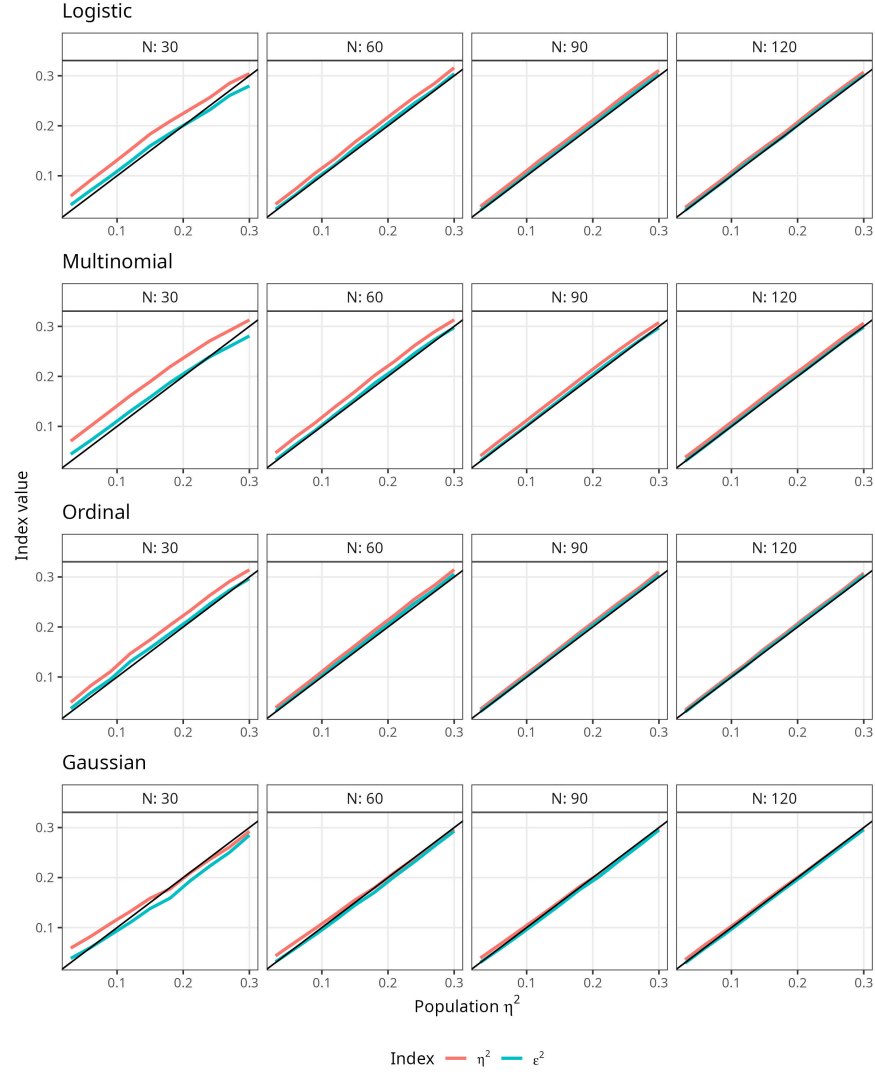


Figure 1: Estimated eta-squared and adjusted epsilon-squared against the population eta-squared, in the supplementary simulation with nonzero nuisance-covariate effects. Black lines represent the population effect size value.

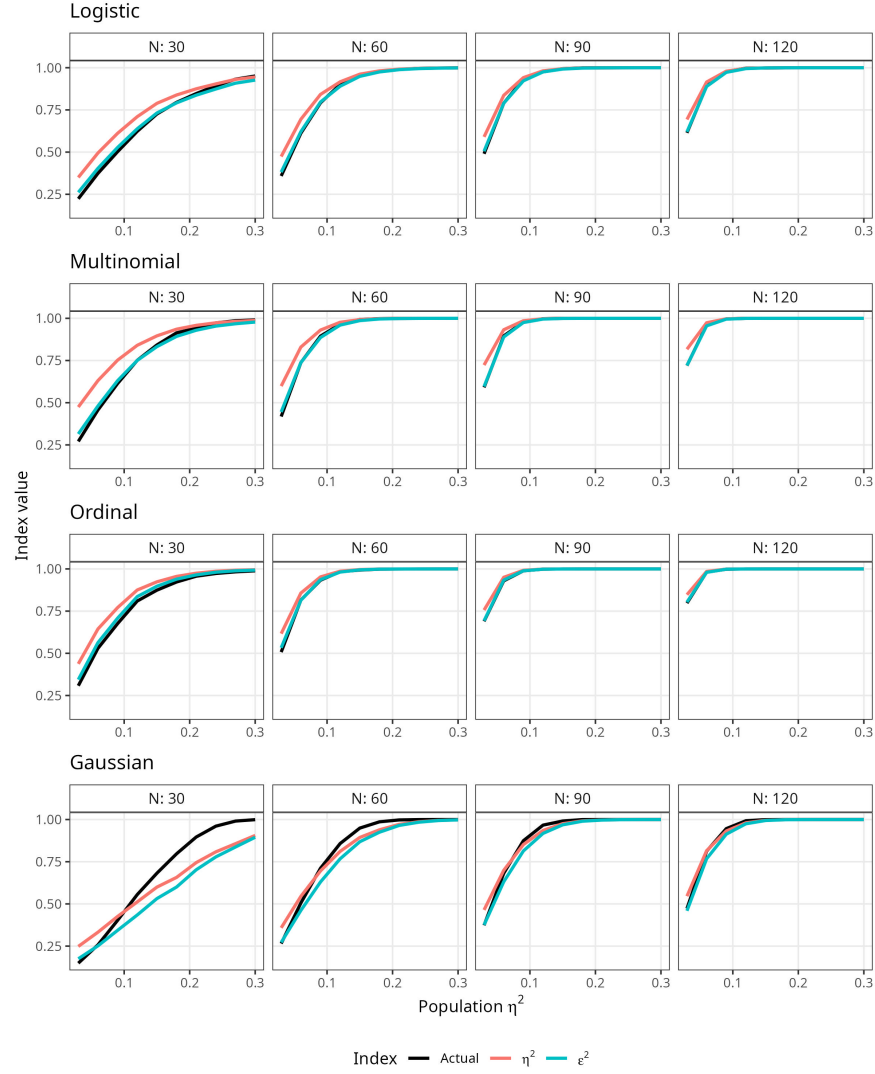


Figure 2: Power estimates for the LR test of the focal predictor based on eta-squared and adjusted epsilon-squared, in the supplementary simulation with nonzero nuisance-covariate effects. Actual represents the empirical power computed in the simulated samples.

Table 5: Accuracy of eta-squared and epsilon-squared, by model and N, comparing the main-text condition (nuisance covariates with zero effect) against the supplementary condition (nuisance covariates at half the focal coefficient), collapsed across population eta-squared, $k = 3$ only.

Model	N	Condition	Bias (Eta2)	Mean Bias (Eta2)	RMSE (Eta2)	Bias (Epsilon2)	Mean Bias (Epsilon2)
Logistic	30	Nuisance = 0	0.032	0.032	0.125	0.009	0.009
Logistic	30	Nuisance = .5x focal	0.024	0.024	0.120	0.002	0.009
Logistic	60	Nuisance = 0	0.015	0.015	0.082	0.003	0.003
Logistic	60	Nuisance = .5x focal	0.016	0.016	0.083	0.004	0.004
Logistic	90	Nuisance = 0	0.010	0.010	0.065	0.002	0.002
Logistic	90	Nuisance = .5x focal	0.011	0.011	0.066	0.003	0.003
Logistic	120	Nuisance = 0	0.007	0.007	0.055	0.001	0.001
Logistic	120	Nuisance = .5x focal	0.007	0.007	0.056	0.001	0.001
Multinomial	30	Nuisance = 0	0.034	0.034	0.095	0.003	0.010
Multinomial	30	Nuisance = .5x focal	0.034	0.034	0.096	0.004	0.010
Multinomial	60	Nuisance = 0	0.020	0.020	0.064	0.004	0.005
Multinomial	60	Nuisance = .5x focal	0.019	0.019	0.064	0.003	0.004
Multinomial	90	Nuisance = 0	0.012	0.012	0.050	0.002	0.002
Multinomial	90	Nuisance = .5x focal	0.012	0.012	0.050	0.002	0.002
Multinomial	120	Nuisance = 0	0.008	0.008	0.042	0.001	0.001
Multinomial	120	Nuisance = .5x focal	0.009	0.009	0.042	0.001	0.001
Ordinal	30	Nuisance = 0	0.026	0.026	0.100	0.010	0.010
Ordinal	30	Nuisance = .5x focal	0.022	0.022	0.099	0.006	0.006
Ordinal	60	Nuisance = 0	0.012	0.012	0.065	0.004	0.004
Ordinal	60	Nuisance = .5x focal	0.012	0.012	0.067	0.004	0.004
Ordinal	90	Nuisance = 0	0.008	0.008	0.051	0.003	0.003
Ordinal	90	Nuisance = .5x focal	0.008	0.008	0.053	0.002	0.002
Ordinal	120	Nuisance = 0	0.006	0.006	0.044	0.002	0.002
Ordinal	120	Nuisance = .5x focal	0.006	0.006	0.045	0.002	0.002
Gaussian	30	Nuisance = 0	0.009	0.013	0.112	-0.016	0.017
Gaussian	30	Nuisance = .5x focal	0.007	0.011	0.099	-0.011	0.013
Gaussian	60	Nuisance = 0	0.004	0.007	0.080	-0.009	0.009
Gaussian	60	Nuisance = .5x focal	0.004	0.005	0.070	-0.005	0.006
Gaussian	90	Nuisance = 0	0.003	0.004	0.065	-0.006	0.006
Gaussian	90	Nuisance = .5x focal	0.002	0.004	0.057	-0.004	0.005
Gaussian	120	Nuisance = 0	0.002	0.003	0.056	-0.005	0.005
Gaussian	120	Nuisance = .5x focal	0.002	0.003	0.049	-0.003	0.003

Power of the single-predictor test, with and without nonzero nuisance effects

Table 6: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, by model and N, comparing the main-text condition (nuisance covariates with zero effect) against the supplementary condition (nuisance covariates at half the focal coefficient), collapsed across population eta-squared, $k = 3$ only.

Model	N	Condition	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Logistic	30	Nuisance = 0	0.058	0.016
Logistic	30	Nuisance = .5x focal	0.060	0.019
Logistic	60	Nuisance = 0	0.027	0.005
Logistic	60	Nuisance = .5x focal	0.028	0.006
Logistic	90	Nuisance = 0	0.016	0.003

Table 6: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, by model and N, comparing the main-text condition (nuisance covariates with zero effect) against the supplementary condition (nuisance covariates at half the focal coefficient), collapsed across population eta-squared, $k = 3$ only. (*continued*)

Model	N	Condition	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Logistic	90	Nuisance = .5x focal	0.016	0.003
Logistic	120	Nuisance = 0	0.011	0.002
Logistic	120	Nuisance = .5x focal	0.011	0.002
Multinomial	30	Nuisance = 0	0.069	0.019
Multinomial	30	Nuisance = .5x focal	0.071	0.018
Multinomial	60	Nuisance = 0	0.032	0.006
Multinomial	60	Nuisance = .5x focal	0.032	0.005
Multinomial	90	Nuisance = 0	0.020	0.002
Multinomial	90	Nuisance = .5x focal	0.018	0.002
Multinomial	120	Nuisance = 0	0.011	0.001
Multinomial	120	Nuisance = .5x focal	0.011	0.001
Ordinal	30	Nuisance = 0	0.044	0.011
Ordinal	30	Nuisance = .5x focal	0.053	0.019
Ordinal	60	Nuisance = 0	0.017	0.002
Ordinal	60	Nuisance = .5x focal	0.018	0.003
Ordinal	90	Nuisance = 0	0.009	0.001
Ordinal	90	Nuisance = .5x focal	0.009	0.001
Ordinal	120	Nuisance = 0	0.005	0.001
Ordinal	120	Nuisance = .5x focal	0.005	0.001
Gaussian	30	Nuisance = 0	0.049	0.033
Gaussian	30	Nuisance = .5x focal	0.099	0.119
Gaussian	60	Nuisance = 0	0.032	0.009
Gaussian	60	Nuisance = .5x focal	0.034	0.042
Gaussian	90	Nuisance = 0	0.022	0.004
Gaussian	90	Nuisance = .5x focal	0.019	0.020
Gaussian	120	Nuisance = 0	0.015	0.002
Gaussian	120	Nuisance = .5x focal	0.011	0.012

Across every model and sample size, accuracy and power under the nonzero-nuisance condition are essentially unchanged from the main-text condition: mean absolute bias in η^2 and ϵ^2 differs by at most a few thousandths, and the direction and approximate magnitude of the ϵ^2 bias-correction pattern described in the main text is preserved. The mean power gap for ϵ^2 is somewhat larger under nonzero nuisance effects (0.017 vs. 0.007 in the main-text condition) but remains small in absolute terms. These results support treating the main-text η^2 simulation design as a reasonable proxy for the more general case in which the other covariates also have non-zero effects.

Study 2: required sample size (data underlying Figure 5)

Table 7: Required sample size for 80% power, omnibus test: simulated vs. closed-form.

Model	Population R2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	312	315
Logistic	0.050	156	157
Logistic	0.075	102	105
Logistic	0.100	77	79

Table 7: Required sample size for 80% power, omnibus test: simulated vs. closed-form. (*continued*)

Model	Population R2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.125	60	63
Logistic	0.150	51	52
Logistic	0.175	42	45
Logistic	0.200	38	39
Multinomial	0.025	246	248
Multinomial	0.050	122	124
Multinomial	0.075	80	83
Multinomial	0.100	59	62
Multinomial	0.125	46	50
Multinomial	0.150	38	41
Multinomial	0.175	32	35
Multinomial	0.200	28	31
Ordinal	0.025	198	198
Ordinal	0.050	98	99
Ordinal	0.075	65	66
Ordinal	0.100	49	50
Ordinal	0.125	39	40
Ordinal	0.150	32	33
Ordinal	0.175	27	28
Ordinal	0.200	24	25

Table 8: Required sample size for 80% power, single-predictor test: simulated vs. closed-form.

Model	Population Eta2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	226	226
Logistic	0.050	114	113
Logistic	0.075	75	75
Logistic	0.100	56	57
Logistic	0.125	44	45
Logistic	0.150	37	38
Logistic	0.175	32	32
Logistic	0.200	29	28
Multinomial	0.025	176	175
Multinomial	0.050	87	88
Multinomial	0.075	57	58
Multinomial	0.100	43	44
Multinomial	0.125	34	35
Multinomial	0.150	28	29
Multinomial	0.175	25	25
Multinomial	0.200	23	22
Ordinal	0.025	144	143
Ordinal	0.050	73	71
Ordinal	0.075	48	48
Ordinal	0.100	36	36
Ordinal	0.125	29	29
Ordinal	0.150	25	24
Ordinal	0.175	21	20
Ordinal	0.200	19	18